Assignment 2

CHE – 321

Chemical Reaction Engineering

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Problems:

P3-6$_A$ The frequency of fireflies flashing and the frequency of crickets chirping as a function of temperature follow. Source: Keith J. Laidler, "Unconventional applications of the Arrhenius law." J. Chem. Educ., 5, 343 (1972). Copyright (c) 1972, American Chemical Society. Reprinted by permission.

For fireflies:			
T (°C)	21.0	25.00	30.0
Flashes/min	9.0	12.16	16.2

For crickets:			
T (°C)	14.2	20.3	27.0
Chirps/min	80	126	200

The running speed of ants and the flight speed of honeybees as a function of temperature are given below. Source: B. Heinrich, *The Hot-Blooded Insects* (Cambridge, MA: Harvard University Press, 1993).

For ants:				
T (°C)	10	20	30	38
V (cm/s)	0.5	2	3.4	6.5

For honeybees:				
T (°C)	25	30	35	40
V (cm/s)	0.7	1.8	3	?

(a) What do the firefly and cricket have in common? What are their differences?
(b) What is the velocity of the honeybee at 40°C? At –5°C?
(c) Nicolas wants to know if the bees, ants, crickets, and fireflies have anything in common. If so, what is it? You may also do a pair-wise comparison.
(d) Would more data help clarify the relationships among frequency, speed, and temperature? If so, in what temperature should the data be obtained? Pick an insect, and explain how you would carry out the experiment to obtain more data. For an alternative to this problem, see CDP3-A_B.
(e) Data on the tenebrionid beetle whose body mass is 3.3 g show that it can push a 35-g ball of dung at 6.5 cm/s at 27° C, 13 cm/s at 37° C, and 18 cm/s at 40° C.
- **(1)** How fast can it push dung at 41.5° C? Source: B. Heinrich. *The Hot-Blooded Insects* (Cambridge, MA: Harvard University Press, 1993). (***Ans.:*** 19.2 cm/s at 41.5° C)

- **(2)** Apply one or more of the six ideas in Preface Table P-4, page xxviii, to this problem.

P2-4$_B$ The exothermic reaction of stillbene (A) to form the economically important trospophene (B) and methane (C), i.e.,

$$A \longrightarrow B + C$$

was carried out adiabatically and the following data recorded:

X	0	0.2	0.4	0.45	0.5	0.6	0.8	0.9
$-r_A$ (mol/dm^3·min)	1.0	1.67	5.0	5.0	5.0	5.0	1.25	0.91

The entering molar flow rate of A was 300 mol/min.

(a) What are the PFR and CSTR volumes necessary to achieve 40% conversion? (V_{PFR} = 72 dm^3, V_{CSTR} = 24 dm^3)
(b) Over what range of conversions would the CSTR and PFR reactor volumes be identical?
(c) What is the maximum conversion that can be achieved in a 105-dm^3 CSTR?
(d) What conversion can be achieved if a 72-dm^3 PFR is followed in series by a 24-dm^3 CSTR?
(e) What conversion can be achieved if a 24-dm^3 CSTR is followed in a series by a 72-dm^3 PFR?
(f) Plot the conversion and rate of reaction as a function of PFR reactor volume up to a volume of 100 dm^3.

Solution:

P.2-4$_b$

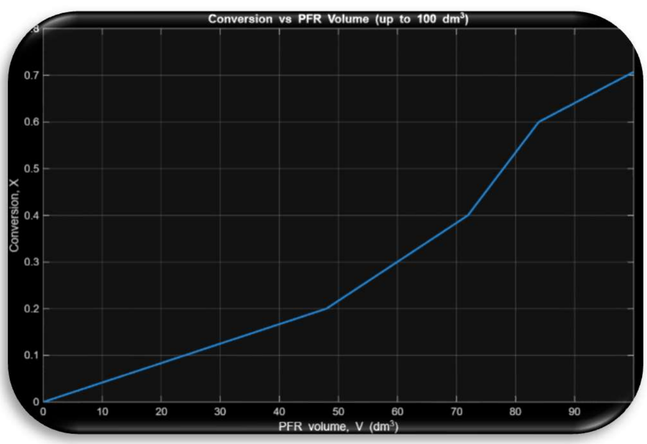
MATLAB Code:

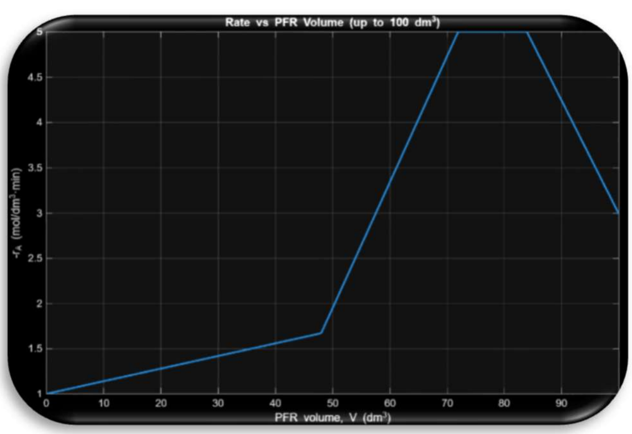
```
%% P2-4B – Graphs only (part f)
FA0 = 300;                        % mol/min

% Table (X, -rA)
Xtab = [0   0.2 0.4 0.45 0.5 0.6 0.8 0.9];
rtab = [1.0 1.67 5.0 5.0  5.0 5.0 1.25 0.91];

% Build Levenspiel cumulative PFR volume from 0 using trapezoids on 1/(-rA)
InvR   = 1./rtab;
Vcum   = FA0 * cumtrapz(Xtab, InvR);      % PFR volume from 0 to each tabulated X

% Create a V-grid up to 100 dm^3 and invert V->X by interpolation
Vmax   = 100;
Vgrid  = linspace(0, Vmax, 401);
XofV   = interp1(Vcum, Xtab, Vgrid, 'linear', 'extrap');
```

```
r_of_V = interp1(Xtab, rtab, XofV, 'linear', 'extrap');


% Plot X vs V

figure; plot(Vgrid, XofV, 'LineWidth', 1.8);

xlabel('PFR volume, V (dm^3)'); ylabel('Conversion, X');

title('Conversion vs PFR Volume (up to 100 dm^3)'); grid on;


% Plot -r_A vs V

figure; plot(Vgrid, r_of_V, 'LineWidth', 1.8);

xlabel('PFR volume, V (dm^3)'); ylabel('-r_A (mol/dm^3·min)');

title('Rate vs PFR Volume (up to 100 dm^3)'); grid on;
```

Answer

Part f:





Problem P2–4B:

Given: $F_{A0} = 300.000\,\text{mol/min}$, adiabatic data as $-r_A$ vs. conversion X:

X	0	0.2	0.4	0.45	0.5	0.6	0.8	0.9
$-r_A$ (mol/dm^3/min)	1.00	1.67	5.00	5.00	5.00	5.00	1.25	0.91

For Levenspiel plots we use $1/(-r_A)$:

X	0	0.2	0.4	0.45	0.5	0.6	0.8	0.9
$1/(-r_A)$	1.000	0.5988	0.2000	0.2000	0.2000	0.2000	0.8000	1.0989

Key design equations

$$V_{\text{PFR}} = F_{A0} \int_{X_1}^{X_2} \frac{dX}{-r_A(X)} \tag{1}$$

$$V_{\text{CSTR}} = F_{A0} \frac{X_{\text{out}} - X_{\text{in}}}{-r_A(X_{\text{out}})} \tag{2}$$

(a) Volumes to reach $X = 0.40$

PFR. Use trapezoids on $[0, 0.4]$:

$$\underbrace{\frac{1 + 0.5988}{2}(0.2)}_{0 \to 0.2} = 0.15988, \qquad \underbrace{\frac{0.5988 + 0.2}{2}(0.2)}_{0.2 \to 0.4} = 0.07988,$$

$$\int_0^{0.4} \frac{dX}{-r_A} = 0.15988 + 0.07988 = 0.23976.$$

Thus

$$\boxed{V_{\text{PFR}} = 300(0.23976) = 71.930\,\text{dm}^3 \approx 72.000\,\text{dm}^3}.$$

CSTR.

$$\boxed{V_{\text{CSTR}} = \frac{300(0.40)}{5.00} = 24.000\,\text{dm}^3}.$$

(b) Range where PFR and CSTR give identical volumes

When $-r_A$ is constant, strip (PFR) area equals rectangle (CSTR) area. From the table $-r_A = 5$ on $[0.40, 0.60]$:

$$\boxed{0.40 \le X \le 0.60}.$$

(c) Max conversion in a $105.000\,\mathrm{dm}^3$ CSTR

Set $V/F_{A0} = 105/300 = 0.35$ in (2) (with $X_{\text{in}} = 0$):

$$0.35 = \frac{X}{-r_A(X)}.$$

On $0.6 \le X \le 0.8$, interpolate

$$-r_A(X) = 5 + \frac{1.25 - 5}{0.2}(X - 0.6) = 16.25 - 18.75X.$$

Solve

$$\frac{X}{16.25 - 18.75X} = 0.35 \;\Rightarrow\; X = 0.35(16.25 - 18.75X) \;\Rightarrow\; \boxed{X \approx 0.752}.$$

(d) $72.000\,\mathrm{dm}^3$ PFR followed by $24.000\,\mathrm{dm}^3$ CSTR

After PFR. $V/F_{A0} = 72/300 = 0.24$. From part (a), $\int_0^{0.4} dX/(-r_A) \approx 0.23976 \approx 0.24$, so $X_1 \simeq 0.40$.

CSTR to X_2.

$$\frac{24}{300} = 0.08 = \frac{X_2 - 0.40}{-r_A(X_2)}, \quad -r_A(X) = 16.25 - 18.75X \;(0.6 < X < 0.8).$$

Thus

$$X_2 - 0.40 = 0.08(16.25 - 18.75X_2) \;\Rightarrow\; 2.5X_2 = 1.70 \;\Rightarrow\; \boxed{X_2 = 0.68}.$$

(e) $24.000\,\mathrm{dm}^3$ CSTR followed by $72.000\,\mathrm{dm}^3$ PFR

CSTR first. $24/300 = 0.08 = X_1/(-r_A(X_1))$. Taking $-r_A = 5$ in the flat region gives $X_1 = \boxed{0.40}$.

PFR next. Need area

$$\frac{72}{300} = 0.24 = \int_{0.40}^{X_2} \frac{dX}{-r_A}.$$

Split:

$$\int_{0.40}^{0.60} \frac{dX}{5} = \frac{0.20}{5} = 0.04, \quad \Rightarrow \quad \int_{0.60}^{X_2} \frac{dX}{-r_A} = 0.20.$$

On $0.60 \to 0.80$, $-r_A = 16.25 - 18.75X$ so

$$\int_{0.60}^{0.80} \frac{dX}{-r_A} = \frac{1}{18.75} \ln\left(\frac{5}{1.25}\right) = \frac{1}{18.75} \ln 4 \approx 0.0739.$$

Remaining area needed: $0.20 - 0.0739 = 0.1261$ from $0.80 \to X_2$. On $0.80 \to 0.90$,

$$-r_A(X) = 1.25 - 3.4(X - 0.8).$$

Hence

$$\int_{0.80}^{X_2} \frac{dX}{-r_A} = -\frac{1}{3.4} \ln\left(\frac{-r_A(X_2)}{1.25}\right) = 0.1261 \Rightarrow \frac{-r_A(X_2)}{1.25} = e^{-0.4287} = 0.651.$$

Solve for X_2 using $1.25 - 3.4(X_2 - 0.8) = 0.8138$:

$$X_2 - 0.8 = \frac{1.25 - 0.8138}{3.4} = 0.128, \qquad \boxed{X_2 \approx 0.93}.$$

P. 3-6A

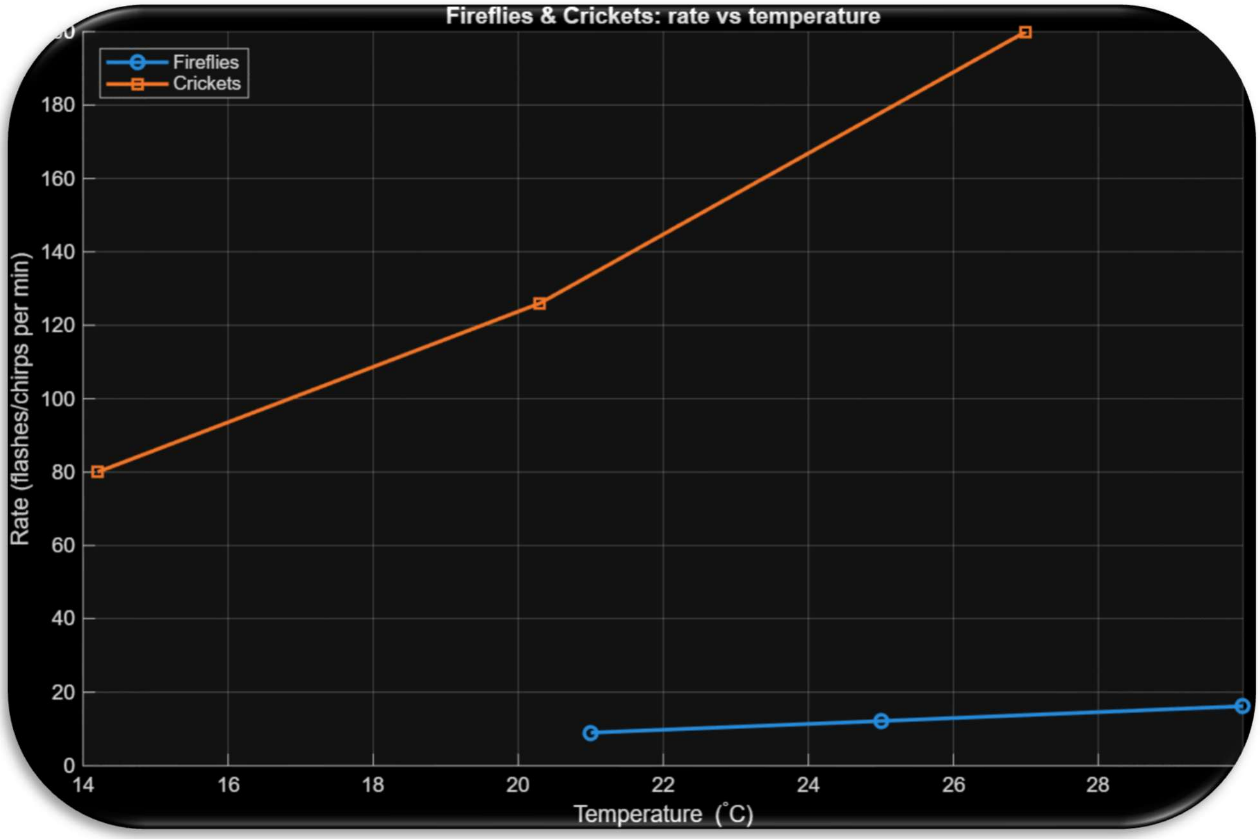
MATLAB code:

```
clear; clc; close all;

R = 8.314;



% Fireflies (flashes/min)
T_ff_C = [21 25 30];
k_ff   = [9.0 12.16 16.2];

% Crickets (chirps/min)
T_ck_C = [14.2 20.3 27.0];
k_ck   = [80 126 200];

% Ants (cm/s)
T_an_C = [10 20 30 38];
v_an   = [0.5 2.0 3.4 6.5];

% Honeybees (cm/s)  (unknown at 40C left blank)
T_be_C = [25 30 35 40];
v_be   = [0.7 1.8 3.0 NaN];

% Beetle (cm/s)
T_bt_C = [27 37 40];
v_bt   = [6.5 13 18];

% Kelvin & Arrhenius abscissa helpers
toK  = @(Tc) Tc + 273.15;
```

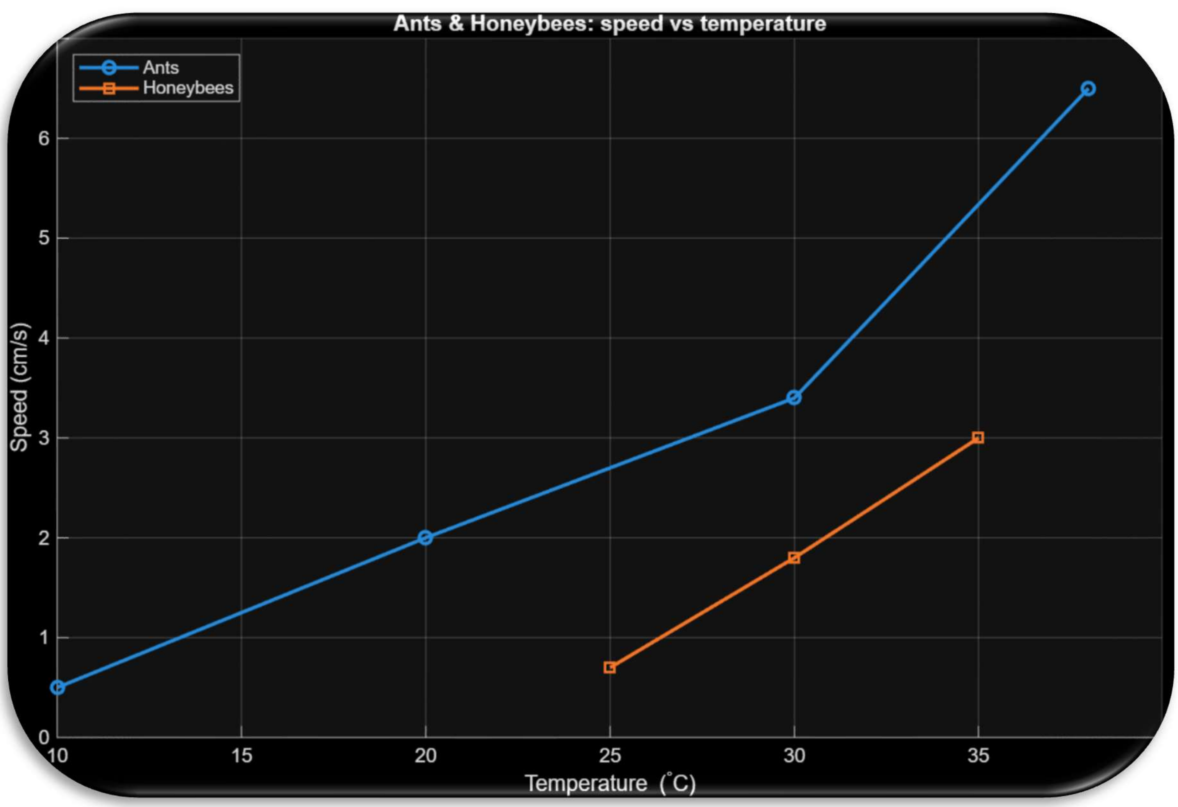
```
invT = @(Tc) 1 ./ toK(Tc);



figure('Name','Rates vs Temperature'); hold on; grid on;
plot(T_ff_C, k_ff, 'o-', 'LineWidth', 1.8);
plot(T_ck_C, k_ck, 's-', 'LineWidth', 1.8);
xlabel('Temperature (^{\circ}C)');
ylabel('Rate (flashes/chirps per min)');
title('Fireflies & Crickets: rate vs temperature');
legend('Fireflies','Crickets','Location','northwest');

figure('Name','Speeds vs Temperature'); hold on; grid on;
plot(T_an_C, v_an, 'o-', 'LineWidth', 1.8);
mask = ~isnan(v_be);
plot(T_be_C(mask), v_be(mask), 's-', 'LineWidth', 1.8);
xlabel('Temperature (^{\circ}C)');
ylabel('Speed (cm/s)');
title('Ants & Honeybees: speed vs temperature');
legend('Ants','Honeybees','Location','northwest');



figure('Name','Fireflies Arrhenius');  arr_plot(T_ff_C, k_ff, 'Fireflies: Arrhenius plot', invT);
figure('Name','Crickets Arrhenius');   arr_plot(T_ck_C, k_ck, 'Crickets: Arrhenius plot', invT);
figure('Name','Ants Arrhenius');       arr_plot(T_an_C, v_an, 'Ants: Arrhenius plot', invT);
figure('Name','Honeybees Arrhenius');  arr_plot(T_be_C(mask), v_be(mask), 'Honeybees: Arrhenius plot', invT);
figure('Name','Beetle Arrhenius');     arr_plot(T_bt_C, v_bt, 'Beetle: Arrhenius plot', invT);
```

```
function arr_plot(Tc, y, ttl, invTfun)

    x = invTfun(Tc);
    yl = log(y);

    hold on; grid on;
    plot(x, yl, 'o', 'LineWidth', 1.8);

    p = polyfit(x, yl, 1);
    xfit = linspace(min(x)*0.98, max(x)*1.02, 200);
    yfit = polyval(p, xfit);
    plot(xfit, yfit, '-', 'LineWidth', 1.5);

    xlabel('1/T (K^{-1})');
    ylabel('ln(rate)');
    title(ttl);
    legend('data','linear fit','Location','best');
end
```

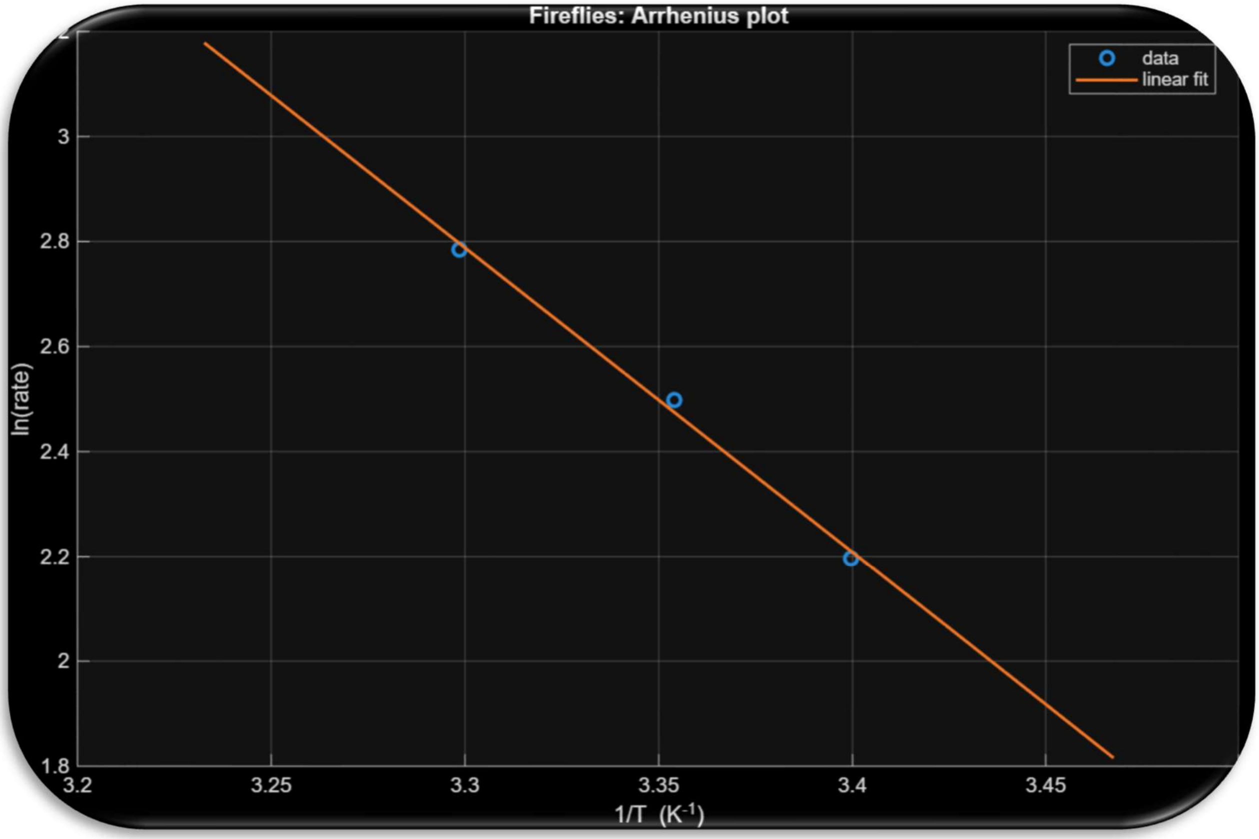
Plots:

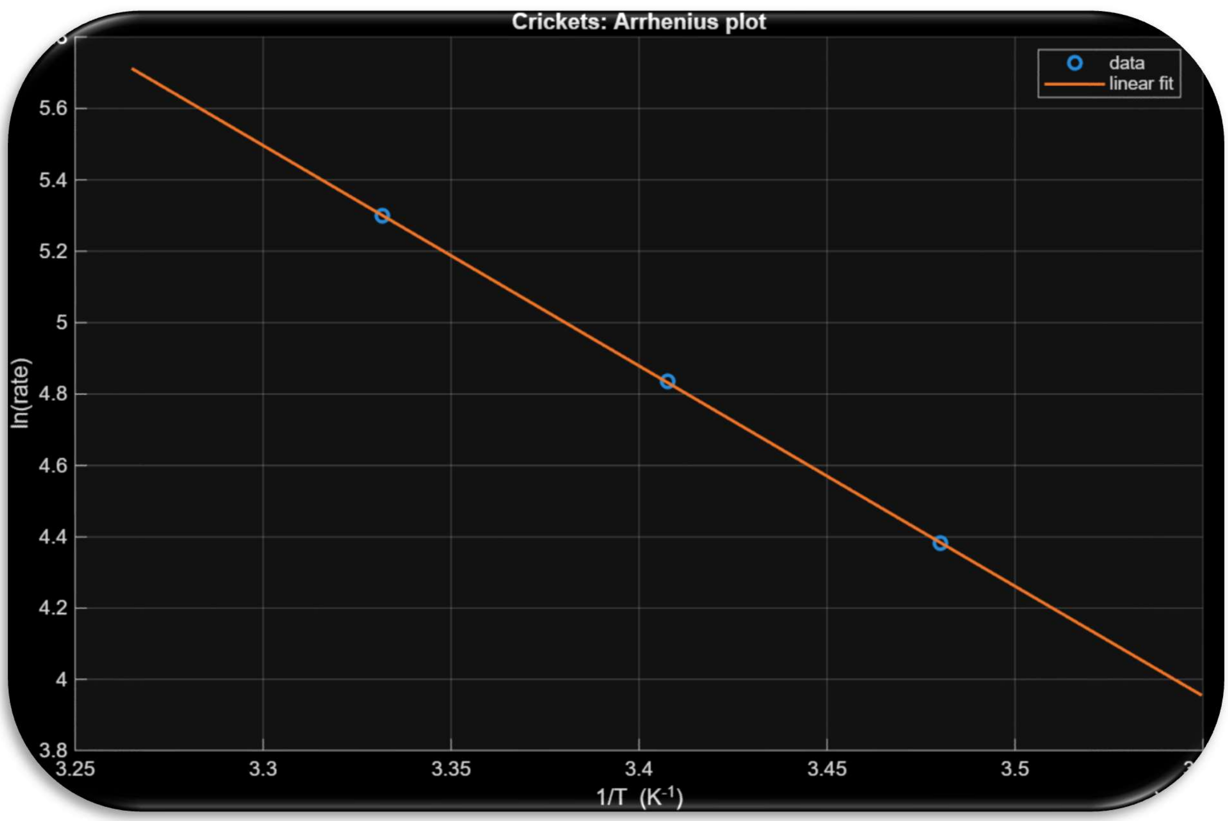


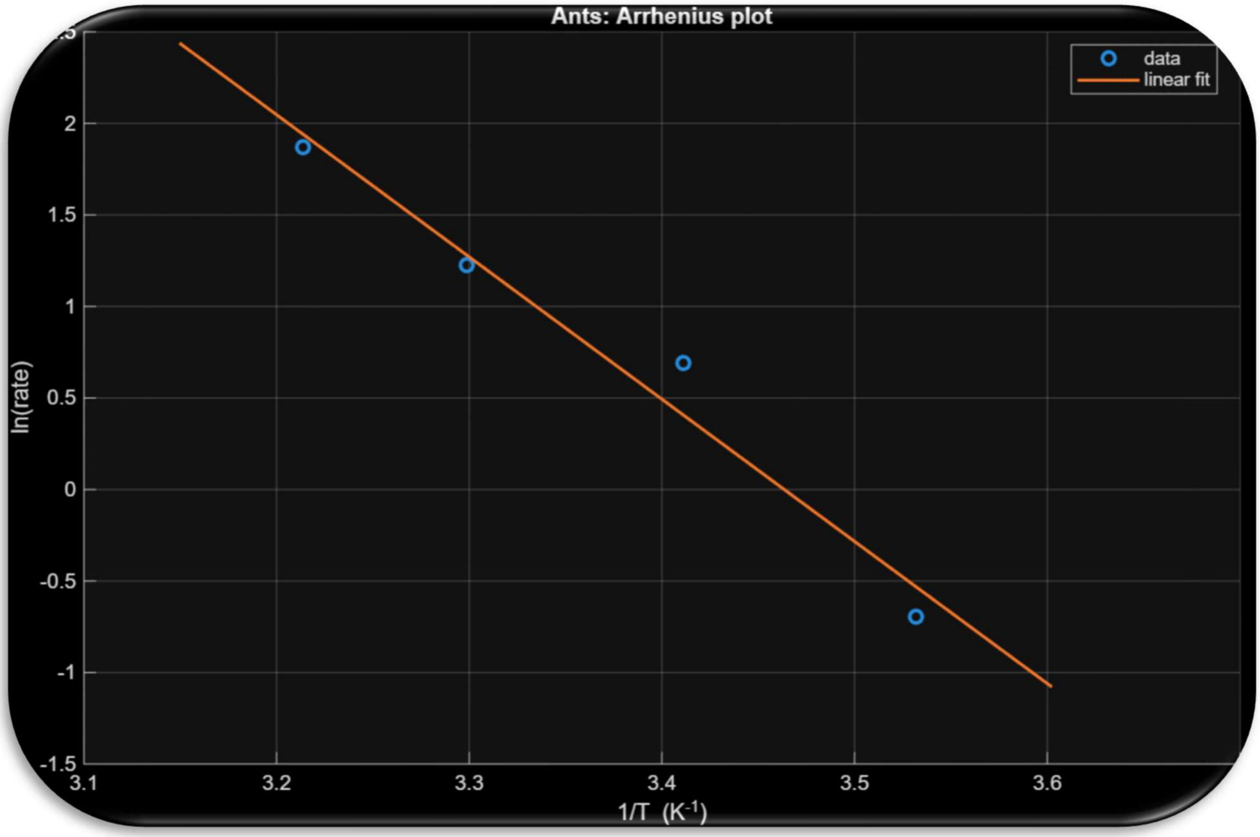
Fireflies & Crickets: rate vs temperature
Fireflies
Crickets
Rate (flashes/chirps per min)
180
160
140
120
100
80
60
40
20
0
14
16
18
20
22
24
26
28
Temperature (˚C)

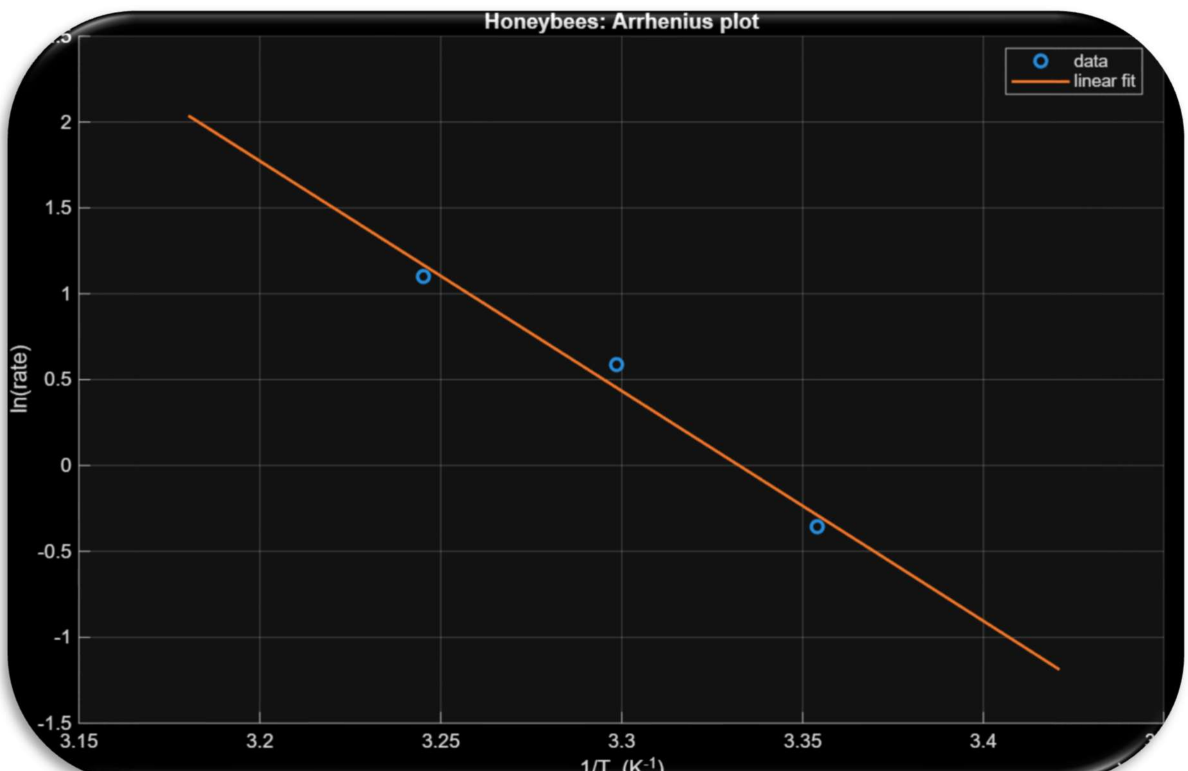


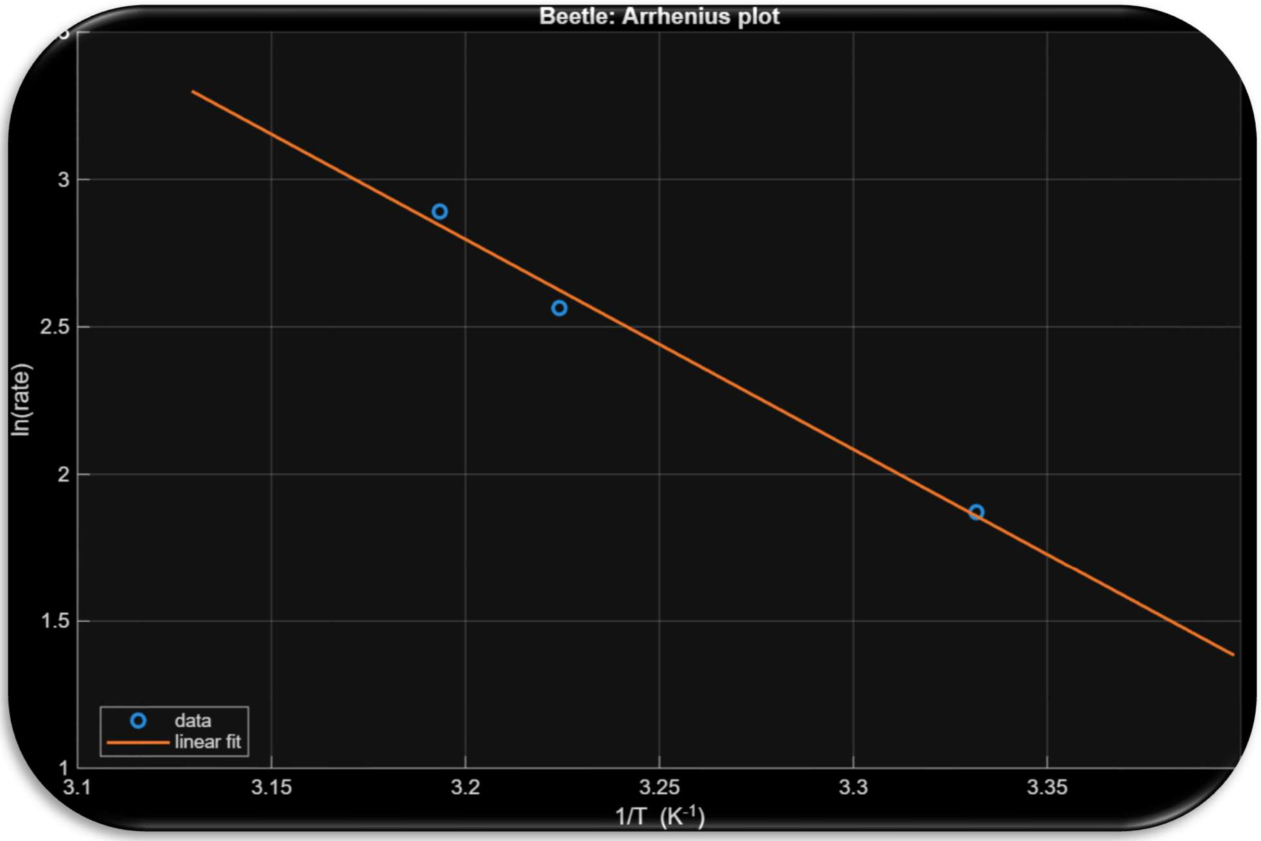
Ants & Honeybees: speed vs temperature
Ants
Honeybees
Speed (cm/s)
6
5
4
3
2
1
0
10
15
20
25
30
35
Temperature (˚C)


Fireflies: Arrhenius plot
data
linear fit
ln(rate)
1/T (K-1)
3
2.8
2.6
2.4
2.2
2
1.8
3.2
3.25
3.3
3.35
3.4
3.45


Crickets: Arrhenius plot
data
linear fit
ln(rate)
1/T (K-1)
5.6
5.4
5.2
5
4.8
4.6
4.4
4.2
4
3.8
3.25
3.3
3.35
3.4
3.45
3.5


Ants: Arrhenius plot
data
linear fit
ln(rate)
1/T (K-1)


Honeybees: Arrhenius plot
data
linear fit
ln(rate)
1/T (K-1)



Beetle: Arrhenius plot
3
2.5
2
1.5
1
ln(rate)
3.1
3.15
3.2
3.25
3.3
3.35
1/T (K-1)
data
linear fit

Problem P3–6A:

Arrhenius model:

$$k(T) = A\,e^{-E_a/(RT)}, \qquad \ln k = \ln A - \frac{E_a}{R}\frac{1}{T}, \qquad E_a = -R\,\frac{\ln(k_2/k_1)}{(1/T_2 - 1/T_1)}.$$

All T in Kelvin: $T[\mathrm{K}] = T[°C] + 273.15$, $R = 8.314\,\mathrm{J\,mol^{-1}\,K^{-1}}$.

Raw data

Fireflies (flashes/min)

$T(°C)$	21	25	30
k	9.0	12.16	16.2

Ants (cm/s)

$T(°C)$	10	20	30	38
v	0.5	2.0	3.4	6.5

Honeybees (cm/s)

$T(°C)$	25	30	35
v	0.7	1.8	3.0

Crickets (chirps/min)

$T(°C)$	14.2	20.3	27.0
k	80	126	200

(a) Firefly vs. cricket: commonality & difference

Both show Arrhenius behavior (linear $\ln k$ vs. $1/T$) with similar activation energies. Two-point estimates using extreme temperatures:

Fireflies (21→30°C).

$$T_1 = 294.15\,\mathrm{K},\ k_1 = 9.0; \quad T_2 = 303.15\,\mathrm{K},\ k_2 = 16.2$$

$$\boxed{E_a = -R\frac{\ln(16.2/9.0)}{(1/303.15 - 1/294.15)} = 48.420\,\mathrm{kJ\,mol^{-1}}}.$$

Crickets (14.2→27.0°C).

$$T_1 = 287.35\,\mathrm{K},\ k_1 = 80; \quad T_2 = 300.15\,\mathrm{K},\ k_2 = 200$$

$$\boxed{E_a = 51.330\,\mathrm{kJ\,mol^{-1}}}.$$

Common: Arrhenius trend, $E_a \sim 50.000\,\mathrm{kJ\,mol^{-1}}$. *Different:* absolute rates and temperature spans.

(b) Honeybee velocity at 40°C and at −5°C

Estimate E_a from $25 \to 35$°C:

$$T_1 = 298.15\,\mathrm{K},\ v_1 = 0.7; \quad T_2 = 308.15\,\mathrm{K},\ v_2 = 3.0$$

$$\boxed{E_a = 111.160\,\mathrm{kJ\,mol^{-1}}}.$$

Predict with $v(T) = v_2 \exp\left[-\frac{E_a}{R}\left(\frac{1}{T} - \frac{1}{T_2}\right)\right]$.

At 40°C ($T = 313.15$ K):

$$v = 3.0 \exp\left[-\frac{111161.84}{8.314}\left(\frac{1}{313.15} - \frac{1}{308.15}\right)\right] = \boxed{6.000\,\mathrm{cm\,s^{-1}}}.$$

At −5°C ($T = 268.15$ K):

$$v = 3.0 \exp\left[-\frac{111161.84}{8.314}\left(\frac{1}{268.15} - \frac{1}{308.15}\right)\right] = \boxed{4.640 \times 10^{-3}\,\mathrm{cm\,s^{-1}}}.$$

(c) Do they have something in common?

All show Arrhenius sensitivity. Quick E_a estimates:

Fireflies $\approx 48.400\,\mathrm{kJ/mol}$, Crickets $\approx 51.300\,\mathrm{kJ/mol}$, Ants (10→38°C) $\boxed{E_a \approx 67.100\,\mathrm{kJ/mol}}$, Honeybee

Thus bees are most temperature-sensitive; fireflies ≈ crickets.

(d) Would more data help?

Yes, especially near physiological limits to check for slope changes. Example (ants): environmental chamber at $5, 10, 15, \ldots, 40$°C, ≥ 10 individuals per T, acclimation ≥ 20 min, run on a photogate track; analyze $\ln v$ vs $1/T$ for slope (E_a) and report mean±SD.

(e1) Beetle speed at 41.5°C

Fit $\ln v$ vs $1/T$ through the three points (27, 37, 40°C) gives

$$\boxed{E_a \approx 59.260\,\mathrm{kJ\,mol^{-1}}}.$$

Using $v = Ae^{-E_a/RT}$ at $T = 314.650\,\mathrm{K}$:

$$\boxed{v(41.5°\mathrm{C}) \approx 19.200\,\mathrm{cm\,s^{-1}}}$$